

PrimeKit: Associativity Defect Diagnosis & Repair — Consolidated Spec (v2)

This document compiles the developments from PrimeKit v0 $\rightarrow$ v0.1 $\rightarrow$ v2 into a single, falsifiable, runnable research/engineering spec.

1. Philosophy in one line

Learn a **prime basis** and **lawful composition operators**; measure **associativity defects**; apply **minimal, costed augmentations** that improve **ternary generalization** and reduce **defect significance**—or else report a **menu-relative closure defect** as a scientific result.

Key slogans:

- **Lawful composition**: operators are admissible by design + enforced by tests.
 - **Closure diagnostics**: associator is the diagnostic; repair actions are constrained and falsifiable.
 - **Irreducibility is okay**: “persistent associator under the menu” is evidence, not failure.
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2. System overview

PrimeKit is organized into four interacting layers:

2.1 Representation layer (Prime Dictionary Learner)

- Learn a parts-based, nonnegative basis $P \in \mathbb{R}_{\geq 0}^{m \times K}$ and sparse activations $a(x) \in \mathbb{R}_{\geq 0}^K$.
- Enforce identifiability and stability via gauge fixing:
 - column normalization of P
 - deterministic column ordering (e.g., by explained mass)
 - sparsity penalties on a
 - carrier / “no-new-direction” checks
- **extra_prime** knob: increment K only when defects demand it and it passes costed veto.

2.2 Composition layer (Prime-space Composition Operators)

Provide a library of admissible operators $\otimes_\alpha : Z \times Z \rightarrow Z$ (where Z is the latent cone or its extension).

Baseline operator families:

- **LRS** (linear receptor sum): $u \otimes v = s(u + v)$
- **DN** (divisive normalization): $u \otimes v = \frac{u+v}{1+\gamma(u+v)}$ (coordinatewise)
- **operator_mixing**: convex mixtures of admissible operators

Admissibility gates (configurable):

- cone-closure (if operating in a cone)
- continuity + empirical Lipschitz/contractivity bounds
- optional: endpoint identity, monotonicity (domain-specific)

2.3 Diagnostics layer (Associator Spectroscopy)

- Evaluate associativity on ternaries by measuring the associator.
- Quantify defect geometry (low-rank / high-rank / spiky structure).
- Calibrate noise and produce a significance statistic and/or p-value.

2.4 Repair layer (Augmentation Controller)

- A controller proposes a discrete set of augmentations, evaluates them against a veto rule, and commits only the best candidate.
- Augmentations are explicitly costed (MDL-ish bits, degrees of freedom, and/or runtime cost).

Augmentation menu (canonical):

- operator_mixing
- context_1d (state/context-dependent operator parameters)
- sparse_interaction (bounded bilinear/sparse correction)
- extra_prime (increase K)
- netting toggle (group-completion / extended cone)

3. Core diagnostic object: the associator defect tensor

3.1 Associator on ternaries

Given latent states $u_i, v_j, w_k \in Z$, a readout $R : Z \rightarrow \mathcal{Y}$, and a metric d on $\mathcal{Y}$:

$$A_{ijk} = d(R(u_i \otimes v_j \otimes w_k), R(u_i \otimes (v_j \otimes w_k)))$$

3.2 Repeats and heteroscedasticity

PrimeKit v2 promotes **repeat-aware** defects:

- collect $A_{ijk}^{(r)}$ for repeats $r = 1..R$
- compute per-triple mean $\bar{A}_{ijk}$ and variance $\hat{\sigma}_{ijk}^2$

This separates:

- within-triple measurement/realization noise
- across-triple structural variation
- across-seed/training stochasticity (optional replicate mode)

3.3 James–Stein shrinkage for variance stabilization (v2)

Per-entry variance estimates can be noisy for small R . We shrink:

$$\hat{\sigma}_{ijk}^2 = (1 - \lambda) \hat{\sigma}_{ijk}^2 + \lambda \sigma_{\text{pooled}}^2.$$

3.4 Weighted global test statistic (v2)

Use inverse-variance weighting:

$$S = \frac{\sum_{ijk} w_{ijk} \bar{A}_{ijk}}{\sqrt{\sum_{ijk} w_{ijk}}}, \quad w_{ijk} = \frac{1}{\hat{\sigma}_{ijk}^2 + \epsilon}.$$

This is a “global defect” statistic with good power when defects are diffuse.

3.5 Defect geometry summaries

On $\bar{A}$, compute:

- Frobenius norm $\|\bar{A}\|_F$
- spike score $\max |\bar{A}| / (\mathbb{E}|\bar{A}| + \epsilon)$
- low-rank energy profile via SVD on the flattening $\bar{A} \in \mathbb{R}^{I \times (JK)}$
- rank-hat: smallest r s.t. $\|\bar{A} - \bar{A}_r\|_F^2 / \|\bar{A}\|_F^2 \leq \eta$

4. Empirical null: wild bootstrap for S (v2)

Analytic nulls are fragile under heteroscedasticity and dependence.

Wild bootstrap procedure:

1. Compute $\bar{A}$ and residuals $e^{(r)} = A^{(r)} - \bar{A}$.
2. Sample Rademacher signs $s_r \in \{-1, +1\}$.
3. Form bootstrap mean $\bar{A}^* = \frac{1}{R} \sum_r s_r e^{(r)}$ (centered at 0).
4. Compute S^* from $\bar{A}^*$ using the same $\tilde{\sigma}^2$.
5. p-value (upper tail): $p = \Pr(S^* \geq S_{obs})$ with add-one smoothing.

Interpretation:

- If p is small, observed defect is unlikely under “noise-only” residual structure.
- This supports menu-relative defect claims with fewer distributional assumptions.

5. Repair ladder: from geometry to candidate generation

PrimeKit's stance:

- Geometry should **inform which candidates to consider**, not dictate a rigid order.
- Selection should be **optimization-based over actual measured improvements**.

Geometry-informed heuristics for proposing candidates:

- **Low-rank defect** → propose operator_mixing, then context_1d
- **High-rank residual** → include extra_prime early
- **Spiky / structured defects** → propose sparse_interaction
- **Domain licenses cancellation** → propose netting toggle

But: all candidates (and some short combos) are evaluated and scored.

6. Controller: optimization-based selection with rollback (v2)

6.1 Proposal interface

A proposal is a callable that:

1. modifies model state (operator/dictionary/augment params)
2. trains briefly and evaluates
3. returns improvements $(\Delta L, \Delta S)$, cost, and a checkpoint token

6.2 Compound score + costs

PrimeKit v2 uses a normalized tradeoff score:

$$\text{score} = \alpha \frac{\Delta L}{\delta L} + (1 - \alpha) \frac{\Delta S}{\delta S} - \gamma \text{cost}.$$

Acceptance:

- default: require $\Delta L \geq 0$ and $\Delta S \geq 0$
- optional early-training grace: allow one step with strong ΔS even if ΔL is small

Rollback discipline:

- Evaluate all candidates from the same baseline snapshot.
- Commit only the best candidate by restoring its returned checkpoint token.

6.3 Composing augmentations

CandidateFactory can propose short combos (length 2) with joint cost:

- operator_mixing + sparse_interaction
 - context_1d + sparse_interaction
 - mixing + context (if lawful)
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7. “Law cards” and LCC join (conceptual integration)

PrimeKit is designed to plug into a Law-Card / LCC (Lawful Closure Composition) stack:

- A **law card** specifies admissible composition along certain subspaces/aspects.
- A **join** ($\bowtie$) composes states only when permitted by the law card.
- Closure diagnostics test whether a macro operation exists that makes Φ sufficient for bracketing.

Even if a full LCC join engine isn’t implemented in v2 scaffolding, the interface is respected:

- composition operators are modular
 - closure tests are attached to admissibility + associator outcomes
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8. CI and invariants: executable philosophy

Make conceptual constraints enforceable:

8.1 Always-on (typical)

- cone closure (for nonnegative latent cones)
- empirical Lipschitz/contractivity bounds
- deterministic gauge fixing checks (dictionary normalization and ordering)

8.2 Profiled (domain-dependent)

- endpoint identity $u \otimes 0 \approx u$
- monotonicity under coordinate order

8.3 Statistical calibration checks

- repeat stability of $\tilde{\sigma}^2$ under resampling
 - bootstrap null sanity: p-values roughly uniform on associative synthetic data
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9. Benchmarks and validation plan

9.1 Synthetic benchmarks (minimum viable)

Four generators:

1. associative (noise-only)
2. low-rank non-associativity
3. high-rank non-associativity
4. spiky non-associativity

Expected controller behavior:

- associative → stop
- low-rank → mixing/context wins on score-per-cost
- high-rank → extra_prime (or combo including it) wins
- spiky → sparse_interaction wins

9.2 Real-data workflow

1. train representation + binary composition
2. evaluate ternary associator with repeats
3. calibrate significance using repeat-aware variance + bootstrap
4. run controller; commit minimal augmentations only if they improve ternary generalization and reduce defect
5. if persistent: report menu-relative defect (rank profile + p-values)

Deliverables:

- closure certificates $\Delta \leq \epsilon$
- augmentation bits/complexity
- law-card sparsity and stability

10. Implementation artifacts produced so far

PrimeKit v0

- Minimal lawful operators (LRS/DN), associator, basic controller, CI tests.

PrimeKit v0.1

- Controller evaluates all candidates (no rigid priority queue)
- compound normalized score (tradeoff between ΔL and ΔS)
- law profiles: optional endpoint identity and monotonicity checks

PrimeKit v2 scaffolding

- heteroscedastic defect denoising + James–Stein variance shrinkage
- inverse-variance weighted S
- wild bootstrap null + p-value
- candidate factory with combo proposals
- controller with rollback + commit tokens

Local artifacts:

- `primekit_v0.zip`
- `primekit_v0_1.zip`
- `primekit_v2_scaffold.zip`

11. What to do next (v2.1 suggestions)

High-leverage improvements:

1. Add **max-type / BH-FDR** tests for localized defects (complements global weighted-S).
2. Use **studentized bootstrap** for better calibration under heteroscedasticity.
3. Make cost multi-objective: (MDL bits, parameters, runtime latency).
4. Add a “law-card harness” to automatically test operator admissibility under domain profiles.

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